

The Maximum Principle for the Heat Equation

The Heat Equation

We consider the heat equation

$$u_t = ku_{xx}, \quad k > 0$$

on the region

$$a \leq x \leq b, \quad 0 \leq t \leq T.$$

Initial condition

$$u(x, 0) = f(x)$$

Boundary conditions

$$u(a, t) = g_1(t), \quad u(b, t) = g_2(t)$$

Parabolic Boundary

The space–time domain is a rectangle in the (x, t) plane.

Interior:

$$a < x < b, \quad 0 < t \leq T$$

Parabolic boundary:

$$t = 0, \quad x = a, \quad x = b$$

The top edge $t = T$ is not part of the boundary for the evolution problem.

Weak Maximum Principle

Maximum Principle

Suppose u satisfies

$$u_t = ku_{xx}$$

in the interior of

$$a \leq x \leq b, \quad 0 \leq t \leq T.$$

Then

$$\max_{a \leq x \leq b, 0 \leq t \leq T} u(x, t)$$

occurs on the parabolic boundary

$$t = 0, \quad x = a, \quad x = b.$$

Similarly for the minimum.

Strong Maximum Principle

Strong Maximum Principle

Let u satisfy

$$u_t = ku_{xx}$$

in a connected region.

If u achieves a maximum at an interior point

$$(x_0, t_0), \quad t_0 > 0$$

then

$$u(x, t) \equiv \text{constant}$$

throughout the connected region.

Thus a nonconstant solution cannot achieve an interior maximum.

Strategy of the Proof

Assume u has an interior maximum.

At a maximum point

$$(x_0, t_0)$$

we have

$$u_x(x_0, t_0) = 0$$

$$u_{xx}(x_0, t_0) \leq 0$$

$$u_t(x_0, t_0) \geq 0$$

But the PDE gives

$$u_t = ku_{xx}.$$

This does not immediately produce a contradiction.

We modify the function slightly.

The εt Trick

Define

$$v(x, t) = u(x, t) - \varepsilon t$$

with $\varepsilon > 0$.

Then

$$v_t = u_t - \varepsilon$$

$$v_{xx} = u_{xx}$$

Substituting into the PDE gives

$$v_t - kv_{xx} = -\varepsilon < 0.$$

Contradiction Argument

Suppose v has an interior maximum at

$$(x_0, t_0).$$

At this point

$$v_x = 0, \quad v_{xx} \leq 0, \quad v_t \geq 0.$$

Therefore

$$v_t - kv_{xx} \geq 0.$$

But from the modified equation

$$v_t - kv_{xx} = -\varepsilon < 0.$$

This is a contradiction.

Conclusion

Therefore v cannot attain an interior maximum.

Hence the maximum of v occurs on the parabolic boundary.

Letting

$$\varepsilon \rightarrow 0$$

shows the same must be true for u .

Thus the maximum principle holds.

Uniqueness of Solutions

Suppose u and v solve the heat equation with the same data.
Define

$$w = u - v.$$

Then

$$w_t = kw_{xx}$$

with zero initial and boundary conditions.

By the maximum principle

$$\max w = 0, \quad \min w = 0.$$

Hence

$$w \equiv 0$$

and therefore

$$u = v$$

Interpretation

The maximum principle expresses the smoothing nature of heat flow.

- ▶ Heat diffuses from hot regions to cold regions.
- ▶ Temperature peaks cannot spontaneously appear.
- ▶ Extreme values are determined entirely by the boundary and initial data.